

Experiment 28: Medical Diagnosis Using Prolog

Aim

To develop a **simple rule-based medical diagnosis system using Prolog** that identifies possible diseases based on a set of symptoms.

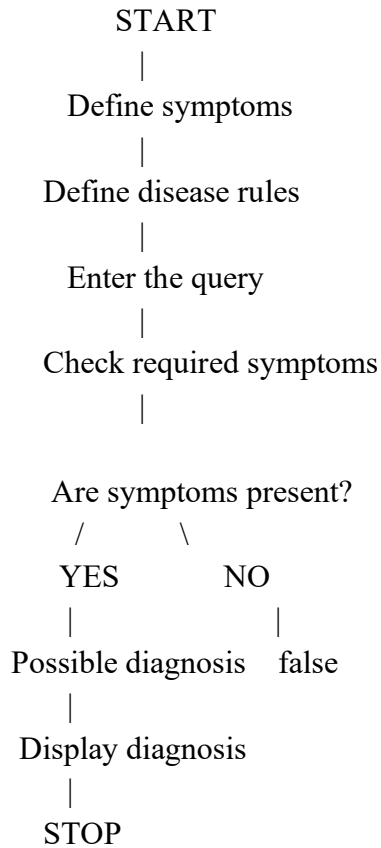
Objective

1. To represent symptoms and diseases using Prolog facts and rules.
2. To understand knowledge representation in Prolog.
3. To use logical rules to identify possible diseases.
4. To execute queries and obtain the corresponding diagnosis.

Algorithm

1. Start the program.
2. Define the symptoms as Prolog facts.
3. Define rules connecting combinations of symptoms with possible diseases.
4. Enter a query for a disease or diagnosis.
5. Prolog checks the required symptoms for that disease.
6. If all required symptoms are available, return true.
7. Otherwise, return false.
8. Use a variable to find all possible diagnoses.
9. Display the result.
10. Stop.

Flowchart



Prolog Code

```
% Medical Diagnosis System
```

```
% Symptoms
```

```
symptom(fever).  
symptom(cough).  
symptom(body_pain).  
symptom(sore_throat).  
symptom(runny_nose).  
symptom(headache).  
symptom(rash).
```

```
% Diagnosis rules
```

```
diagnosis(flu) :-  
    symptom(fever),  
    symptom(cough),  
    symptom(body_pain).
```

```
diagnosis(common_cold) :-  
    symptom(cough),  
    symptom(runny_nose),  
    symptom(sore_throat).
```

```
diagnosis(measles) :-  
    symptom(fever),  
    symptom(rash),  
    symptom(headache).
```

```
% Rule to display diagnosis
```

```
medical_diagnosis(Disease) :-  
    diagnosis(Disease).
```

Sample Output

Query 1: Check for Flu

```
?- diagnosis(flu).
```

Query 2: Check for Common Cold

```
?- diagnosis(common_cold).
```

Query 3: Check for Measles

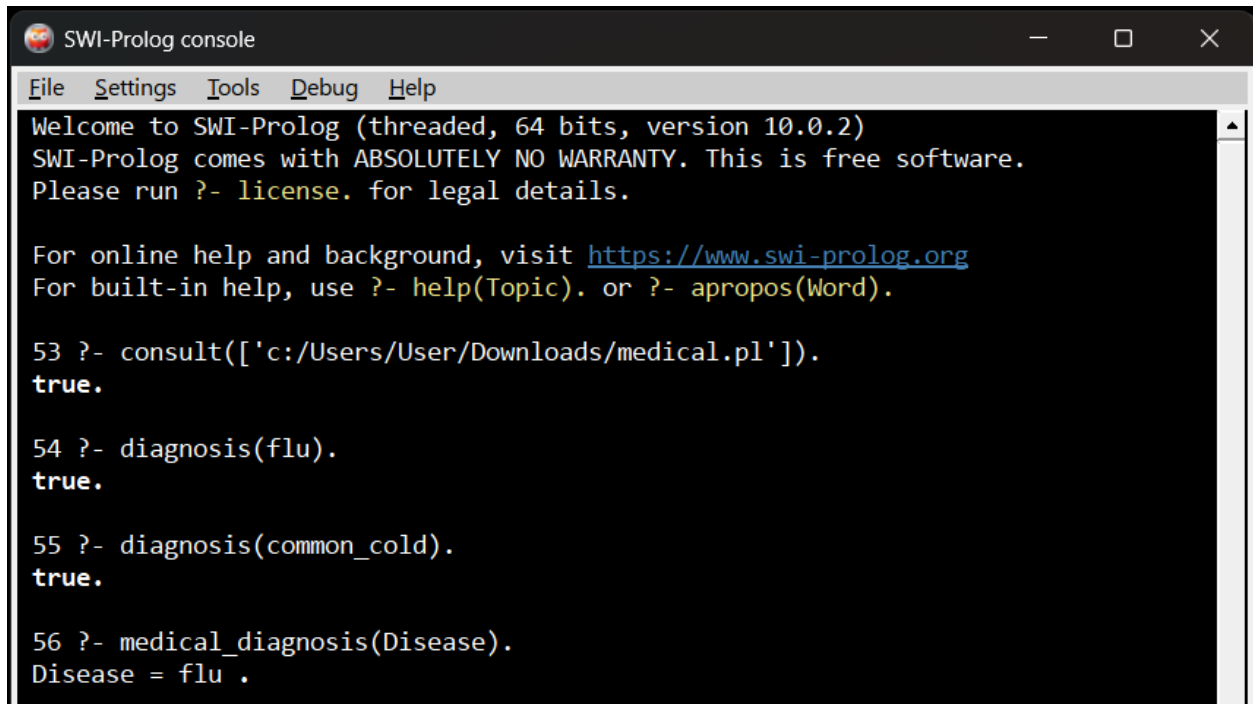
```
?- diagnosis(measles).
```

Query 4: Find all possible diagnoses

```
?- medical_diagnosis(Disease).
```

Query 5: Check an unknown disease

?- diagnosis(malaria).

A screenshot of the SWI-Prolog console window. The window has a title bar with the SWI-Prolog logo and the text "SWI-Prolog console". Below the title bar is a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The main area of the window is a black terminal with white text. It shows the following text:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

53 ?- consult(['c:/Users/User/Downloads/medical.pl']).
true.

54 ?- diagnosis(flu).
true.

55 ?- diagnosis(common_cold).
true.

56 ?- medical_diagnosis(Disease).
Disease = flu .
```

Result

The **medical diagnosis system** was successfully implemented using **Prolog**. The program correctly identifies possible diseases based on predefined symptoms and rules.

Conclusion

Thus, the experiment demonstrates how **Prolog** can be used to build a simple rule-based **medical diagnosis system**. Facts represent symptoms, while logical rules are used to determine possible diseases.

Note: This is an academic demonstration only and should not be used for actual medical diagnosis or treatment.